# Shashank Jaiswal

Full-Stack Developer | shashank3021195@gmail.com | Raipur, Chhattisgarh, India
GitHub: github.com/shashankjai

## EDUCATION

**National Institute of Technology Raipur - B.Tech in Information Technology**
2023 – 2027 (Expected) | CGPA: 9.15/10

## PROJECTS

### SkillSetu (Full-Stack Development)

An AI-powered peer-to-peer skill exchange platform that helps users discover learning partners, exchange skills, and coordinate learning sessions.

- AI-powered skill matching using embeddings and cosine similarity
- Gemini-powered RAG assistant for personalized learning roadmaps
- JWT-based authentication
- Real-time messaging and notifications with Socket.IO
- Session scheduling and WebRTC video calls

Tech: React.js, Tailwind CSS, Node.js, Express.js, MongoDB, Gemini API, RAG, JWT, Socket.IO, WebRTC

### Rembrio (AI / Full-Stack Development)

An AI-powered study assistant that helps users learn from PDFs with grounded answers, page-level citations, quizzes, and personalized study plans.

- PDF question answering with page-level citations
- RAG using page-aware chunking and Gemini embeddings
- MongoDB Atlas Vector Search with cosine-similarity fallback
- AI quizzes, study plans, and progress analytics
- Authentication, request validation, and rate limiting
- Automated tests for APIs, authorization, and citations

Tech: React.js, Node.js, Express.js, MongoDB, Gemini API, RAG, Vector Search, JWT, Zod

### IoTSentinel SOC (Machine Learning / Cybersecurity)

A deep-learning project focused on detecting benign and malicious IoT activity using aggregated STRACE system-call features.

- Classifies IoT activity as benign or malicious from aggregated STRACE system-call features
- Deep-learning detection model trained on system-call behaviour
- FastAPI backend serving predictions to a React SOC dashboard
- Data preparation and feature handling with Pandas and NumPy

Tech: Python, Scikit-learn, FastAPI, React.js, Pandas, NumPy

### Deep Packet Inspection Engine (C++ / Systems / Networking)

A C++ deep packet inspection engine for analyzing network traffic, identifying flows, classifying applications, and applying configurable traffic-filtering rules.

- Reads and analyzes network traffic from PCAP captures
- Identifies flows and classifies applications, including TLS traffic
- Configurable rules for filtering and blocking traffic
- Multithreaded packet processing in C++ with the STL

Tech: C++, STL, Multithreading, TCP/IP, TLS, PCAP

## SKILLS

Programming Languages: C++, Python, JavaScript, SQL
Frontend: React.js, Vite, Tailwind CSS, HTML, CSS, Recharts
Backend: Node.js, Express.js, FastAPI, REST APIs, Socket.IO, WebRTC
Databases & Authentication: MongoDB, MongoDB Atlas Vector Search, JWT, Zod
AI / ML: Scikit-learn, Pandas, NumPy, Retrieval-Augmented Generation (RAG), Google Gemini API, Embeddings
Developer Tools: VS Code, Git, GitHub, Postman
Core Subjects: Data Structures and Algorithms, Object-Oriented Programming, Database Management Systems, Computer Networks, Operating Systems

## ACHIEVEMENTS

- 2000+ - DSA problems solved
- 1839 - LeetCode rating
- 3-Star - CodeChef, max rating 1629

- 1422 - Codeforces rating
- 1696 - GeeksforGeeks rating
- 2nd Runner-Up - Xcelerate 2026, IIIT Naya Raipur
- Rank 338 - CodeChef Starters 189 (Global)
- Rank 1681 - LeetCode Weekly Contest 504 (Global)
- Rank 1264 - Codeforces Round 1050 (Div. 4)
- 16th Rank - Uttar Pradesh, Class 10 boards

## EXPERIENCE

**Senior Designer - SAHAYOG – The Student Wellbeing Club, NIT Raipur (2023 – 2026)**

- Led a team of 5 designers to create posters, banners and digital creatives for 10+ campus events.
- Mentored junior designers and maintained consistent branding and visual communication.